DHAIRYA CHHABRA

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 3+ years of experience designing, building, and shipping end-to-end web applications and cloud-native backend services across education, retail, and enterprise platforms. Skilled in **Java, Spring Boot, Python, FastAPI, Django, Node.js, Express.js, TypeScript, React.js, Next.js, REST APIs, PostgreSQL, MongoDB, Redis, Apache Kafka, AWS, Docker, Kubernetes, and CI/CD**. Comfortable owning features from responsive React interfaces through distributed microservices, event-driven pipelines, and production monitoring, with a focus on performance, reliability, testing, and clean, maintainable code.

TECHNICAL SKILLS

Programming Languages: Java | TypeScript | JavaScript (ES6+) | Python | SQL | C++
Frontend: React.js | Next.js | Redux | HTML5 | CSS3 | Responsive Web Design
Backend Engineering: Spring Boot | Node.js | Express.js | FastAPI | Django | REST APIs | Microservices | Distributed Systems | Event-Driven Architecture
Databases & Messaging: PostgreSQL | MongoDB | MySQL | Redis | Apache Kafka
Cloud & DevOps: AWS (Lambda, S3, CloudWatch) | Docker | Kubernetes | Jenkins | GitHub Actions | CI/CD
Testing & Observability: JUnit | PyTest | Jest | Postman | Grafana | CloudWatch | OpenTelemetry
Practices & Tools: Agile | Scrum | SDLC | TDD | Git | GitHub | Jira | Linux

PROFESSIONAL EXPERIENCE

San Diego State University **San Diego, CA**
Graduate Software Engineer **Sep 2024 - Present**

- Led a 6-person team to design and deploy a distributed AI chatbot on AWS, building Python backend services with Flask and FastAPI and exposing them through REST APIs to improve internal-portal information-retrieval accuracy by 20%.
- Designed Java-based automated grading services and wrote comprehensive JUnit and PyTest suites under a test-driven-development workflow, cutting manual grading effort by 40% across 200+ students.
- Optimized PostgreSQL query performance by profiling slow queries and adding composite indexes on high-frequency filter columns, accelerating lookups roughly 12x (about 150 ms to 12 ms) by eliminating full-table scans.
- Containerized backend services with Docker and deployed on AWS, using CloudWatch for centralized logging and metrics to keep the chatbot and grading systems reliable in production.
- Mentored 200+ students on debugging, refactoring, Git workflows, and core software-engineering practices, improving overall code quality and project outcomes.

Wordcorn **San Jose, CA**
Software Engineer Intern **May 2025 - Aug 2025**

- Built end-to-end full-stack features — authentication, user progress tracking, and personalized learning workflows — with a React.js and TypeScript frontend backed by Python REST APIs.
- Developed scalable backend services in Python to power progress tracking and content-recommendation logic for the education platform.
- Launched the platform's first revenue stream by integrating Stripe end to end, implementing idempotent, webhook-driven payment flows in Node.js that required no manual reconciliation.
- Improved global performance by introducing edge caching and CDN optimization for media assets, cutting page-load times by 40% and origin infrastructure costs by 25%.
- Set up observability with AWS CloudWatch and Grafana dashboards to monitor API latency, system health, and error rates, reducing mean time to resolution by 40%.

AFour Technologies **Pune, India**
Trainee Software Engineer **Mar 2023 - Mar 2024**

- Designed and built Java Spring Boot microservices using an asynchronous, event-driven architecture with message queues, scaling ingestion throughput 4x by decoupling background processing from the request path.
- Built additional backend services in Python and JavaScript on AWS and reduced distributed-microservice API response latency by 20% through profiling and targeted optimization.
- Established CI/CD pipelines for the Spring Boot services with Jenkins and Docker, adding automated test gates to catch regressions before release and reduce manual deployment overhead.

- Instrumented services with distributed tracing and OpenTelemetry and centralized logs and metrics in CloudWatch and Grafana dashboards, speeding production root-cause isolation 3x.
- Hardened backend authentication and the PostgreSQL data layer with parameterized queries and stateless token verification, cutting unauthorized-access attempts by 95%.
- Redesigned multi-region state synchronization around eventual-consistency guarantees to cut cross-region message-delivery latency by 60%, working in an Agile, cross-functional team.

Digilytics Systems LLP **Pune, India**
Software Engineer Intern **Dec 2021 - May 2022**

- Built a no-code website builder with a React frontend and REST APIs, enabling 500+ small businesses to launch webpages and lifting client sales by about 20%.
- Implemented Redis caching and optimized PostgreSQL storage schemas to reduce repeated database queries under peak load and keep response times low.
- Containerized the application and set up Docker-based CI/CD, accelerating release cycles 2x by removing error-prone manual deployment steps.
- Wrote Jest test suites and followed Git-based version control to maintain code quality across the team.

PROJECTS

BudgetWise — Full-Stack Expense Manager | Tech Stack: React.js, Node.js, Express, MongoDB, Redux, REST APIs

- Built a full-stack expense-management application with a React and Redux frontend and a Node.js/Express backend, helping users track budgets and trim monthly spending by about 10%.
- Designed RESTful APIs and modeled data in MongoDB to support real-time updates, category breakdowns, and smooth data flow.
- Implemented a responsive React UI with client-side state management for a consistent experience across devices.

RootLens AI — Distributed Observability Pipeline | Tech Stack: Python, Go, FastAPI, Kafka, Docker

- Architected a scalable observability pipeline with FastAPI and Go microservices, streaming high-volume logs and traces through Kafka to detect anomalies and correlate cross-service failures.
- Containerized the services with Docker for reproducible local and cloud deployment, reducing MTTR in simulated incident scenarios.

EDUCATION

Master of Science in Computer Science | San Diego State University, San Diego, CA | May 2026 | GPA: 3.73
Bachelor of Engineering in Computer Engineering | Savitribai Phule Pune University, Pune, India | May 2023 | GPA: 3.94